

Corridor, Not Factory: Trade Reorientation and the Missing Investment Response in Kazakhstan, 2022–2025

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Abstract

A large, sector-specific rise in demand for a tradeable good is expected to induce domestic investment—import substitution, value-chain entry, new capacity. After February 2022 Kazakhstan received such a shock: with Russia’s direct trade disrupted, part of it was rerouted through neighbouring economies. We measure the shock at the product level (HS6, 2018–2025), estimate what Kazakhstan retains, and ask whether it induced investment. In 29 product lines exports to Russia rose roughly tenfold, breaking in 2022m5 (sup- F of 561); imports of the same goods rose too, and a difference-in-differences against control lines puts the outbound increase at about $3.5\times$. Kazakhstan re-exports these goods with no more than a trade-and-logistics margin: propagated through the input–output table, the retained margin implies domestic value added of about a tenth of a dollar per dollar rerouted (5–9%), against roughly three-quarters for a dollar of domestic manufacturing. And there is no investment response: in deal data from three commercial databases, dealmaking in manufacturing, transport and logistics runs at the same rate after 2022 as before, 2022 is the weakest year, and no transaction over 2015–2025 falls in the product lines the reorientation flows through. We read the null through a three-gate decomposition—irreversibility, market access (a

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customs union makes trade a substitute for production), and capital-market institutions. Two within-country comparisons point to the irreversibility gate: captive state capital also abstained, and the same institutions built real capacity for a *durable* auto-demand shock but none for the transitory one. Kazakhstan operates as a corridor, not a factory—the trade statistics move by an order of magnitude, the domestic economy barely registers it.

Keywords: trade reorientation, intermediated trade, investment under uncertainty, irreversibility, industrialization, value added, institutional voids, transit economies, Kazakhstan. **JEL:** F14, F15, E22, D25, O14, O24, O33, P33.

1 Introduction

When demand for a country’s tradeable output rises sharply and persistently, the textbook expectation is that supply follows: firms expand, new entrants appear, and—if the country was importing the good—some of the demand is met domestically. This logic underlies a large part of development economics, from infant-industry and import-substitution arguments to modern work on global-value-chain entry and the investment effects of trade shocks. Its premise is that a demand signal, once clear enough and durable enough, is acted on by investment.

After February 2022 Kazakhstan received an unusually clean version of such a signal. The disruption of direct trade between Russia and its principal partners produced a visible redirection of trade through third countries. Chupilkin et al. [2026] document the pattern: a sharp fall in EU exports to Russia alongside a matching rise in EU exports to Armenia, Kazakhstan and the Kyrgyz Republic, with product-level and mirror-statistics evidence of onward movement to Russia; Chupilkin et al. [2025] quantify the substitution. For Kazakhstan the reorientation is concentrated in a well-defined set of technology-intensive products and is large: at the monthly level, exports to Russia in these lines rose roughly tenfold. The routes, the demand and the products are all observable in the trade data.

This paper asks what Kazakhstan did with that signal. We proceed in three steps.

First, we measure the shock. Using UN Comtrade at HS6, annual for 2018–2025 and monthly for 2019–2025, we identify from the data a “surge basket” of 29 product lines in which both the inbound flow (West and China) and the outflow to Russia at least doubled after 2022. Both break in 2022m5 (outbound sup- F of 561, inbound 24), with a flat pre-trend and a plateau after six months; Armenia and the Kyrgyz Republic show the same 2022 break with no comparable domestic reform. A difference-in-differences against control lines puts the outbound increase at $\gamma = 2.44$ ($3.5\times$ in levels), matched by a pre-specified priority-list version and by a randomisation-inference benchmark; a placebo that treats the largest civilian lines runs the other way (they contracted after 2022).

Second, we measure what Kazakhstan retains. On the incremental flow to Russia (about \$479m over 2022–2025), the unit-value evidence shows no sign of transformation—no systematic weight gain, and a mark-up no larger than the freight Kazakhstan pays to bring the goods in. Bounding the retained trade-and-logistics margin at 6–12% and propagating it through Kazakhstan’s input-output table (OECD ICIO), the domestic value added per dollar rerouted is about **5–9%**, against roughly three-quarters for a dollar of domestic manufacturing output. Customs duty on the incremental flow is well under 1% of customs revenue, and the flow is 0.2–0.3% of GDP.

Third, we ask whether the shock induced investment—and find that it did not. We assemble deal-level data on mergers, acquisitions, and private-equity and venture transactions in Kazakhstan for 2015–2025 from three commercial databases (Capital IQ, PitchBook, Preqin), with FactSet and Dealroom extracts specified in Appendix A for cross-validation in the final draft, and supplemented by the transaction record of Kazakhstan’s state investment corporation. Transactions in manufacturing of tradeable goods, transport and logistics run at 7.3 per year in 2015–2021 and 7.5 per year in 2022–2025; 2022 is the weakest year in the series. Reported deal value rises, but is concentrated in ownership transfers of existing or distressed assets—two of the largest are nationalisations. Across the whole 2015–2025 window there is *no* transaction in the specific product lines that dominate the reorientation.

We interpret the non-response with a framework in which the supply response to the shock is the product of three gates: an *irreversibility* gate (does anyone build capacity for a shock this transitory and policy-contingent, given the option to serve the demand through trade instead?), a *market-access* gate (is domestic transformation required to reach the final market?), and an *institutional* gate (can the projects that clear the first two be financed?). The gate that is easy to overlook is the market-access one: because Kazakhstan is in a customs union with the destination, trade is a close substitute for production, so the demand shock is largely absorbed by the trade channel and transmits weakly to the investment decision. The unit-value evidence bears on this directly: Kazakhstan does not, in the median line, charge more for the good than it pays for it, so building is dominated before the option value of waiting is even considered. Two within-country comparisons then point to the irreversibility gate as independently binding: the state investment corporation, which faces no financing or exit constraint, also did not invest in the reorientation, and the same institutions in the same years produced real capacity for a *durable* demand shock (vehicle assembly) while producing none for the transitory one. The institutional gate explains a separate fact—why the durable adjacent opportunities the shock reveals, in logistics and machinery import substitution, go unfinanced by private capital.

Contribution. The paper makes three contributions. (i) Its primary contribution is empirical: to our knowledge it is the first host-economy incidence analysis of the post-2022 trade rerouting, extending the existing literature—which establishes that rerouting happened and its scale—to what the intermediary retains, and to 2025. It shows that a large, well-identified, sector-specific tradeable-demand shock produced no detectable investment response, a fact that speaks to when trade shocks induce a domestic supply response and when they do not [Juhász, 2018, Bloom et al., 2016, Juhász et al., 2024, Gopinath and Neiman, 2014]. (ii) It organizes the two standard readings of a missing supply response—investment under un-

certainty and irreversibility [Dixit and Pindyck, 1994], and institutional voids in the capital market [Khanna and Palepu, 1997, Rajan and Zingales, 1998]—into a single decomposition, adding a customs-union market-access gate under which trade substitutes for production, and uses two within-country comparisons to argue the irreversibility gate is the binding one here. The decomposition is an organizing device rather than an estimated model. (iii) Methodologically, it combines public trade data, an input–output table, and multi-source commercial deal data into a replicable measure of what a transit economy captures, cross-checking the deal counts across databases rather than relying on one.

The rest of the paper is organised as follows. Section 2 describes the setting; Section 3 sets out the framework; Section 4 the data, including the deal-database construction; Section 5 measures the reorientation; Section 6 what Kazakhstan retains; Section 7 the (non-)investment response; Section 8 asks which gate binds; Section 9 the moderators and external validity; Section 10 where investment would pay off; and Section 11 concludes.

2 Setting

Kazakhstan is a founding member of the Eurasian Economic Union, which removes the customs frontier between Kazakhstan and Russia: goods that clear customs on entry to Kazakhstan move onward to Russia without a further border, and a supply from Kazakhstan to Russia is treated as intra-union. Kazakhstan is also the land bridge between China and Russia and, via the Trans-Caspian corridor, between China and Europe. These features make it a natural conduit for trade that can no longer move directly, and they mean that acting as a conduit requires no transformation of the goods.

Several things other than the reorientation changed in Kazakhstan around 2022, and each is a potential confound. The “New Kazakhstan” reform programme associated with the presidency of Kassym-Jomart Tokayev—an outward-oriented agenda that gathered pace in 2022 with a June constitutional referendum and a November snap election—would raise trade and investment generally and gradually; we show below that what we observe is instead concentrated in a narrow set of products and discontinuous in 2022m4, that a civilian control basket moves far less, and that Armenia and the Kyrgyz Republic show the same break with no comparable reform. Three further shocks bear specifically on the *investment* null and cannot be differenced away: the January 2022 unrest and state of emergency (“Qandy Qantar”), in the first month of the post-period; the secondary-sanctions and compliance exposure that the rerouting itself created, which deters exactly the foreign acquirers who might have answered the shock; and the 2022–2025 wave of nationalisations, which both crowds the state investment pipeline and signals ownership risk to private entrants. We

return to these where they matter, in Sections 7 and 8; they make the investment null *over-*determined rather than fragile, but they also mean we cannot cleanly attribute it to the trade shock alone.

3 A framework: when a demand shock builds capacity

Consider a potential entrant deciding whether to build capacity to serve a newly enlarged demand for a tradeable good that the country currently imports. The good can be re-exported to the final market without domestic transformation—this is the relevant case for a member of a customs union with the destination. The entrant has two ways to serve the demand. It can *trade*: buy the good abroad and resell it into the final market, earning a margin μ_T per unit, with no sunk cost and the ability to stop costlessly if demand falls. Or it can *produce*: sink an irreversible cost I to build capacity, then earn a margin μ_P per unit on domestic output for as long as it operates, with the capacity worth only scrap if the demand disappears.

Let the enlarged demand support quantity Q per period and persist each period with probability ρ (so expected duration $1/(1 - \rho)$), and let r be the discount rate. Trading is always weakly profitable while the demand lasts, with present value $V_T = \frac{\mu_T Q}{1 - \rho/(1 + r)}$. Building is profitable only if

$$\underbrace{\frac{\mu_P Q}{1 - \rho/(1 + r)}}_{\text{PV of production margin}} - I > \underbrace{\Omega(\sigma, \rho)}_{\text{option value of waiting}} + V_T, \quad (1)$$

where Ω is the value of the option to defer, increasing in the uncertainty σ around the shock's size and duration and decreasing in ρ [Dixit and Pindyck, 1994]. Four observations follow.

(i) *The irreversibility gate.* The right-hand side of (1) is large exactly when the shock is transitory (low ρ), uncertain (high σ , e.g. because it is policy-contingent), and the technology is sunk-cost-intensive (high I). When any of these is extreme, no entrant builds— $R \approx 0$ —regardless of the other conditions. This is a statement about *whether anyone builds*, and it does not depend on who the entrant is or how deep the capital market is.

(ii) *Trade is the outside option, not just “wait”.* Because the good is re-exportable, the alternative to building is not idleness but V_T : the entrant can serve the demand through trade indefinitely. The demand shock raises the return to *both* margins. When the shock is transitory and μ_T is available at low cost, (1) fails and the supply response is trade rather than production. A corollary: if $\mu_P \leq \mu_T$ —if Kazakhstan cannot even charge more for a transformed good than it pays for the input—then building is dominated before Ω is added.

The unit-value evidence in Section 6 is a direct measurement of this parameter.

(iii) *The market-access gate.* If serving the final market instead *required* domestic transformation—binding rules of origin, local-content rules, or simply a customs border that a pure re-export cannot cross—then $V_T \rightarrow 0$ and (1) is far more likely to hold. Membership of a customs union with the destination opens this gate and is what makes trade a viable substitute for production. The same logic runs through the imported-input intensity of the would-be domestic producer: where domestic production would itself lean heavily on imported components—the roundabout structure of Gopinath and Neiman [2014]—the value a local plant adds over a pure re-export is smaller still, and the transformation margin $\mu_P - \mu_T$ narrower. The gate also governs *dynamic* gains. de Souza et al. [2026] show, using Brazilian firm-to-firm technology-transfer records, that a foreign supplier facing a lower tariff shifts from *transferring* technology into the market toward *exporting* goods to it, and that learning from imported goods is less efficient than learning from a transfer—so import liberalisation raises static welfare but slows diffusion and productivity growth. An open market-access gate is the limiting case: with a pure re-export, the foreign firm has no reason to transfer anything, and the host forgoes not only the static value added of Section 6 but the learning that would come with a plant.

(iv) *The institutional gate.* Suppose (1) *does* hold for some project—a durable opportunity with modest sunk costs. It is still realised only if the entrant can finance I and, for an outside equity investor, exit. Where the capital market lacks growth equity and a functioning exit, projects that clear the irreversibility gate are not funded; the capital that does move is captive—retained earnings, or a state or development-finance investor—and is allocated by criteria other than R [Khanna and Palepu, 1997, Rajan and Zingales, 1998].

Collecting the three gates, the investment response to the shock is

$$\begin{aligned}
 R = & \underbrace{f(\rho, \sigma^{-1}, I^{-1})}_{\text{irreversibility}} \times \underbrace{h(\text{transformation required})}_{\text{market access}} \\
 & \times \underbrace{g(\text{financial depth, exit, private/state mix})}_{\text{institutions}}.
 \end{aligned} \tag{2}$$

The structure is multiplicative: a closed gate anywhere drives R to zero, and the gates bear on different questions—the first two on whether production is ever the right way to serve the demand, the third on whether the projects that pass get built. Sections 5–8 show that for Kazakhstan the irreversibility and market-access gates are both adverse, so $R \approx 0$; Section 9 and Section 10 show that the institutional gate explains a separate fact—the durable adjacent opportunities the shock reveals go unfinanced.

A mediation restatement. The same structure can be put in mediation terms; we do so as an interpretive device, not as a model we estimate—the data below have no firm-level panel linking the trade response to the investment response. Let the demand shock X bear on the investment response Y through a direct path $X \rightarrow Y$ and an indirect path $X \rightarrow M \rightarrow Y$, where the mediator M is the volume of re-export activity the shock generates. The shock raises M ; higher M substitutes for domestic capacity, so it works against Y . When the market-access gate is open ($h = 1$, no transformation required), trade is a close substitute and nearly all of the shock is carried by M : the channel that would raise Y is barely opened, and the net effect on investment is close to nil. When the gate is closed ($h < 1$), M absorbs less of the shock and part of X bears on Y directly, with that residual governed by the irreversibility parameters (ρ, σ, I)—whether the un-absorbed demand is large and durable enough to clear (1). Kazakhstan is the polar case: $h = 1$, and the unit-value wedge of Section 6 is a rough estimate of the margin on the traded path, which is no higher than the cost of the input—so little is left for the direct path even before the irreversibility gate is applied.

4 Data

Trade. UN Comtrade, HS 6-digit. Annual 2018–2025 for magnitudes; monthly 2019m1–2025m12 (authenticated API) for break timing and the event study. Kazakhstan stopped reporting monthly to Comtrade after February 2024, so the monthly series uses 2019–2023 and 2025; 2024 is covered by the annual data and by mirror flows. We use Kazakhstan’s reported exports to Russia (complete) and, for the inbound side, a partner-reported (mirror) measure: the sum of exports to Kazakhstan reported by the EU-27, the United Kingdom, the United States, Japan, Korea, Switzerland, Norway (“Western”) and *China*, because Kazakhstan’s own import reports are incomplete for 2020–2022. China supplies about two-thirds of this inbound value and its flow to Kazakhstan was large and rising well before 2022; we therefore also track the Western component on its own, since that is the part that moves with the reorientation. Mirror data overstate Kazakh absorption for goods declared for Kazakhstan and then moved onward, a bias we return to in Sections 5.3 and 6 [Fisman and Wei, 2004]. Neighbour comparisons (Armenia, the Kyrgyz Republic) and the economy-wide imports-by-chapter series used in Section 10 are from the same source.

Product sets. We take 75 HS6 lines: the 50 codes on the February-2024 EU/US/UK/JP *List of Common High Priority Items* (technology-intensive, potentially dual-use) plus 25 clearly civilian control codes. The **surge basket** is defined from the data as the 29 lines in

which the mean inbound flow (West + China) and the mean outflow to Russia both at least doubled from 2019–2021 to 2022–2023 and exceed modest post-period levels; 24 of the 29 are on the priority list. Because the basket is selected on the post-period outcome, we treat the priority list as a pre-specified alternative throughout and benchmark the difference-in-differences against randomly drawn baskets of the same size (Section 5.3).

Input–output. OECD Inter-Country Input–Output tables, 2023 edition, Kazakhstan block, 45 ISIC rev. 4 industries, 2019. We form the domestic technical-coefficient matrix A_d from the Kazakhstan-on-Kazakhstan intermediate block, the domestic Leontief inverse $L_d = (I - A_d)^{-1}$, and the direct domestic value-added coefficients v_d ; the domestic value-added multiplier for sector j is $(v_d' L_d)_j$.

Deal data. Deal-level records of mergers, acquisitions, private-equity buyouts and growth transactions, and venture rounds involving Kazakhstan-domiciled targets, 2015–2025. The core extract combines Capital IQ, PitchBook and Preqin ($n \approx 500$ after de-duplication), classified to sectors and flagged for greenfield versus ownership transfer. For the final draft we reconstruct the same universe from FactSet and Dealroom as a cross-validation step; the exact query specification for all five databases (geography filter, deal types, date range, industry taxonomy, extraction date) is in Appendix A, and Table 3 reports the coverage of the three analysed here. All are commercial but widely available under academic licence, so the extract is replicable. The state investment corporation’s project register—project, sector, region, financing year and cost—is a public disclosure of the fund (QIC/Baiterek); the replication package pins the release date of the register used here and lists the fields drawn from it.

Macro. World Bank World Development Indicators.

5 The demand shock

5.1 Magnitudes

Table 1 reports the surge basket. Kazakhstan’s exports to Russia in these lines were \$7–17m per year in 2018–2021; they were \$128m in 2022 and \$119–145m in 2023–2025—a rise of roughly an order of magnitude, and the sharpest, cleanest part of the picture. The inbound flow (West + China) ran near \$440m per year before 2022 and roughly \$0.8–2.4bn after, but that series is dominated by China, whose exports to Kazakhstan were already large and

Table 1: Surge-basket trade, USD million per year. Inbound is a partner-reported (mirror) measure. China’s 2024 figure is large and preliminary; its 2025 figure is not yet reported, so the 2025 West+China total is the Western component only.

	2018	2019	2020	2021	2022	2023	2024	2025
West + China → Kazakhstan (mirror)	394	427	483	470	837	1,373	2,360	363
<i>of which Western</i>	162	162	168	184	337	444	419	363
Kazakhstan → Russia	17	12	7	8	128	145	119	133

rising before the war (and are only partly reported for 2024–2025). The *Western* component of the inbound flow—the part aligned with the reorientation—rose from about \$170m per year to \$340–440m, a factor of about 2.3.

5.2 Timing and the event study

Monthly, the outbound series (Kazakhstan’s exports to Russia) breaks sharply in 2022m5 (Bai–Perron, 95% CI 2022m4–2022m6; sup- F of 561 against critical values in the low tens), following the March-2022 sanctions with a short lag and preceding every political-calendar date. The inbound series (West + China) breaks in the same month (95% CI 2022m4–2022m6; sup- F of 24), with a second, smaller break in late 2023 as enforcement tightened; the Western component alone breaks in 2022m5 much more sharply (sup- F of 241). On the *annual* panel the outbound and Western-inbound breaks are large (sup- F of 151 and 142), but the West-plus-China inbound break is not significant (sup- F of 4), because the large pre-existing China flow swamps the 2022 movement at annual frequency—one more reason the reorientation is best seen in the outbound series and the Western inbound. The sup- F statistics come from a mean-shift test on a persistent series and are evidence of a break, not a cardinal measure of its size. Figure 1 plots the series; Figure 2 plots a normalised profile, $\text{asinh}(y_t) = \alpha + \sum_{k \neq -1} \beta_k \mathbf{1}[t - t_0 = k] + \varepsilon_t$ with $t_0 = 2022\text{m}3$ (an aggregate time series, not a difference-in-differences; it has no control group and the bands are wide). Both series are flat before 2022, climb over the first half-year, and plateau—exports to Russia by roughly +3 asinh points (more than a tenfold rise in level), the inbound flow by about +1. That the break precedes the June referendum, that the response is confined to a narrow product set, and that Armenia and the Kyrgyz Republic show the same 2022 break with no comparable reform (Section 5.3) are what separate the reorientation from the “New Kazakhstan” agenda.

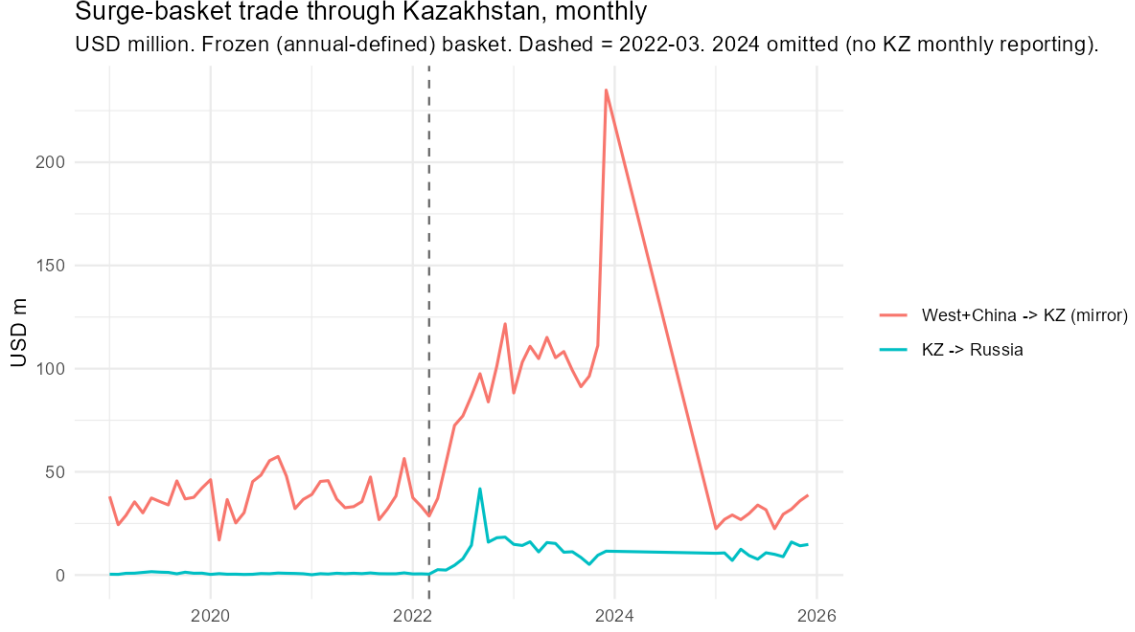


Figure 1: Surge-basket trade through Kazakhstan, monthly. 2024 omitted (no Kazakhstan monthly reporting). Dashed line: March 2022.

5.3 Difference-in-differences

Specification and identifying assumptions. On the annual panel we estimate $\text{asinh}(y_{it}) = \gamma (\text{treated}_i \times \text{post}_t) + \mu_i + \lambda_t + \varepsilon_{it}$ with HS6 and year fixed effects and standard errors clustered by HS6 (75 lines, 8 years). Identification requires that, absent the shock, treated and control lines would have followed parallel paths (a joint test of the three pre-period interaction coefficients does not reject: $p = 0.42$ for the inbound outcome, $p = 0.78$ for the outbound), that there is no anticipation (the reference period is 2022m2; results are similar dropping 2022), and that the surge and control baskets do not contaminate each other. Two features warrant care. First, the surge basket is *selected on the post-period outcome* (lines where both legs at least doubled), which mechanically inflates γ ; we benchmark against randomly drawn baskets of the same size. Second, treatment switches on once at a common date, so cluster-robust standard errors on the HS6 dimension may understate serial correlation; we report wild-cluster-bootstrap p -values alongside.

Results. Table 2 reports γ for the surge basket, the pre-specified priority list, and a placebo. On the surge basket, exports to Russia rise by $\gamma = 2.44$ ($p = 0.013$; wild-bootstrap $p = 0.010$; randomisation-inference $p = 0.009$, i.e. a random basket of 29 lines produces a coefficient this large under one time in a hundred), and a Poisson specification in levels puts the increase at $3.5\times$ ($p = 0.02$). The inbound flow (West + China) rises by $\gamma = 2.10$

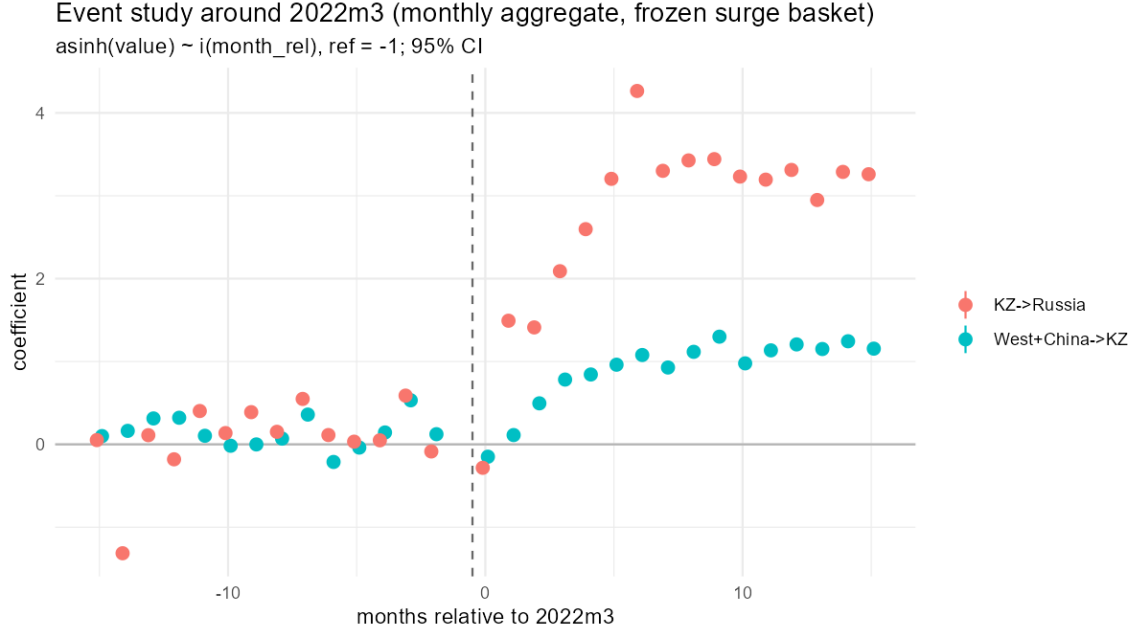


Figure 2: Normalised monthly profile of the surge-basket aggregate around 2022m3 (asinh value, coefficients on event-time indicators relative to 2022m2). This is a single aggregate time series with no control group, not a difference-in-differences event study; it is shown to display the timing and shape of the shift, and the interval bands are correspondingly wide.

($p = 0.05$; $2.4\times$ in levels, $p < 10^{-4}$); the Western component alone by 1.73 ($p = 0.08$) and Kazakhstan’s own reported imports by 1.88 ($p = 0.004$). The pre-specified priority list gives a very similar picture ($\gamma = 2.88$ outbound, 2.12 inbound, both $p < 0.01$), which matters because that list is not selected on the outcome. A placebo that assigns fake treatment to the largest civilian lines gives $\gamma = -0.95$ ($p = 0.001$): the big consumer-goods lines *contracted* after 2022, as the tenge depreciation and a general import compression would predict, while the surge basket rose—so the placebo does not deliver a clean zero, but its sign is the opposite of the treatment effect and it sharpens rather than muddies the contrast. The mirror-gap outcome (partner-reported inflow minus Kazakh-reported imports, a standard onward-movement proxy) does not move significantly ($\gamma = 0.54$, $p = 0.81$); we do not lean on it.

The reform confound. Three things separate the reorientation from the outward-oriented policy agenda of 2022. First, the civilian control basket’s *inbound* flow shows no 2022 break at all (sup- F of 0.3), and while its *exports to Russia* do break (sup- F of 34, $p < 10^{-4}$)—a general Russia-trade effect—the surge-basket outbound break is four to five times larger, and the placebo difference-in-differences runs the wrong way. Second, the outbound break is dated 2022m5, before the June referendum and the November election. Third, and most

Table 2: Difference-in-differences, γ on treated $\times$ post, asinh outcomes, HS6 + year FE, SE clustered by HS6 ($N = 600$). The surge basket is selected on the inbound (West+China) and outbound flows; the priority list (50 codes) is pre-specified. Wild-cluster bootstrap and randomisation inference give the same significance classification.

Treated set	Exports to Russia	Inbound (W+China)	Inbound (Western)	KZ-reported imports
Surge basket (29 HS6)	2.44 (0.96)**	2.10 (1.06)*	1.73 (0.98)	1.88 (0.63)**
Priority list, pre-specified (50 HS6)	2.88 (0.74)***	2.12 (0.71)**	1.93 (0.67)**	2.94 (0.53)***
Placebo (largest civilian lines)	—	−0.95 (0.25)***	—	—

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$ (cluster-robust). Poisson in levels: surge basket $3.5\times$ outbound ($p = 0.02$), $2.4\times$ inbound ($p < 10^{-4}$).

directly: Armenia’s surge-basket exports to Russia rose from \$7–15m per year to \$71m in 2022 and \$93m in 2023 (sup- $F = 10$, $p = 0.01$), and the Kyrgyz Republic’s from \$1–7m to \$12m and \$30m (sup- $F = 13$, $p = 0.004$), on the same timing—neither country having a “New Kazakhstan” reform. The direction is also asymmetric: in from sanctioning economies, out to Russia.

6 What Kazakhstan retains

6.1 The retained trade margin

We measure what Kazakhstan retains on the *incremental* flow—exports to Russia above the pre-2022 baseline, about \$479m cumulatively over 2022–2025—since that is the part the reorientation drove. Against the incremental Western inbound of about \$887m, the incremental flow-through is roughly one half; the rest is domestic use, onward flow to other destinations, and measurement in the import series, which we cannot separate.

The unit-value evidence is genuinely ambiguous and worth stating carefully. Comparing the price Kazakhstan charges Russia (f.o.b.) with the price it *records paying* on import (c.i.f., which includes freight into landlocked Kazakhstan), the median ratio across matched HS6 $\times$ period cells is **0.73** (annual, $n = 112$; 0.70 monthly): Kazakhstan books these goods out at a lower recorded price than it books them in. Comparing instead with the price the *exporter* reports (f.o.b. at the partner’s border, so on the same basis as the outbound price), the median ratio is **1.6** (annual; 1.5 monthly): on a like-for-like basis Kazakhstan does mark the goods up, by an amount that includes the freight it paid to get them in. Neither ratio is distinctive to the reorientation—the civilian control basket gives 0.59 and 0.94 on the two bases—and the pass-through regression is weak on both (a slope of 0.06–0.48, significant only in the annual f.o.b. specification). What the wedge does *not* show, on any basis, is the

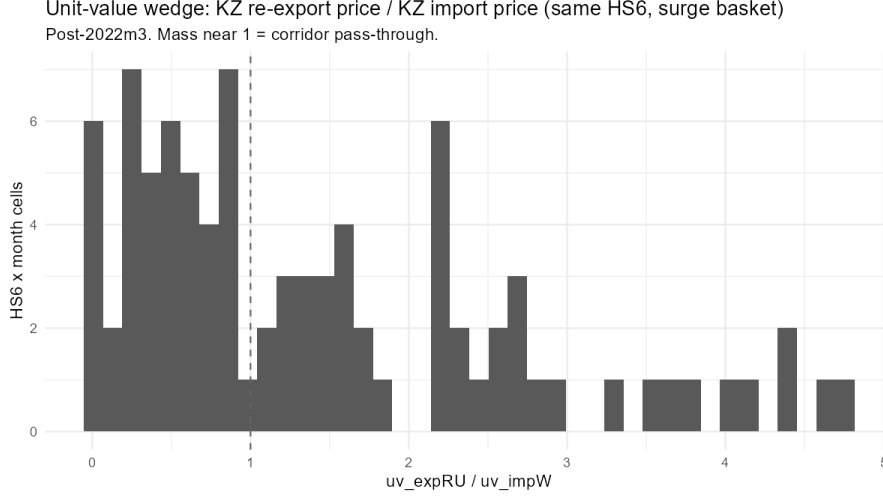


Figure 3: Unit-value wedge: Kazakhstan re-export price over the exporter-reported inbound price (f.o.b. basis), same HS6, surge basket, post-2022. Wide dispersion; no systematic weight gain.

large mark-up or weight gain that domestic transformation would produce. The retained margin is a trade-and-logistics margin, which we bound next.

6.2 Input–output propagation

Let m be the trade-and-freight margin retained on a transiting good as a share of its value, $\bar{v}^{TT}$ the mean domestic value-added multiplier for Kazakhstan’s trade, transport and warehousing sectors, and $\bar{v}^M$ that for manufacturing. Domestic value added per dollar of gross rerouted flow is $m \bar{v}^{TT}$; per dollar of domestic manufacturing output it is $\bar{v}^M$ [Koopman et al., 2014, Johnson and Noguera, 2012]. From the OECD ICIO Kazakhstan block, $\bar{v}^{TT} = 0.79$ and $\bar{v}^M = 0.76$ —close enough that the propagation step does little work here: the result is essentially the retained margin m against the full domestic value $\bar{v}^M$, and the input–output structure mainly confirms that neither trade nor manufacturing has unusual import leakage in Kazakhstan.

We bound m from below by the Kazakhstan national-accounts convention for trade, insurance and freight on goods trade, 6–14%, and from above by the matched-cell f.o.b. gross margin of about 34%—after subtracting the inbound freight Kazakhstan pays on the re-exported quantity (roughly 15% of value into a landlocked economy), that upper bound falls to about 12%. The two approaches therefore agree on a narrow range: $m \approx 6\text{--}12\%$, midpoint about 10%. Domestic value added per dollar rerouted is $m \bar{v}^{TT} \approx 5\text{--}9\%$, midpoint about 8%, against **76%** per dollar of domestic manufacturing output—so a rerouted dollar creates on the order of a tenth as much domestic value as a produced one. Of the \$479m in incremental exports to Russia after 2022, Kazakhstan retains roughly **\$23–45m** as domestic value added.

The conclusion is not sensitive to the manufacturing multiplier: even at $\bar{v}^M = 0.40$ the ratio is about one to five.

6.3 Fiscal and macro

The reorientation-attributable flow—incremental exports to Russia over the pre-2022 baseline—is about \$479m cumulatively over 2022–2025 (\$107–134m per year). Customs duty on it, at an EAEU common external tariff of 4–8% on the share formally cleared in Kazakhstan, is on the order of **\$22m over four years** (about \$5m per year); net import VAT is close to zero because the onward supply to Russia is intra-union and zero-rated. Kazakhstan’s annual customs revenue is of the order of \$4–5bn, so the reorientation adds well under 1%. At the macro level the flow is 0.2–0.3% of GDP; the 2022 current-account surplus, the reserve accumulation of 2023–2025 and the rise in services value added are an order of magnitude too large to be driven by it.

7 The (non-)investment response

If the demand shock were seeding domestic value addition, we would expect a rise in investment in the sectors the flow passes through. We find none—though, with only four post-2022 years, the deal-count test can rule out a large response, not a modest one.

Dealmaking in manufacturing of tradeable goods, transport and logistics, and wholesale distribution runs at **7.3 per year** in 2015–2021 and **7.5 per year** in 2022–2025 (a Poisson rate ratio of 1.03, 95% CI 0.63–1.65); 2022 itself is the weakest year in the series (Figure 4). Dropping the 2022 trough, 2023–2025 runs at 9.0 per year (rate ratio 1.24, 95% CI 0.74–2.01). The test has 80% power only against an increase of roughly 80% or more, so it excludes a surge in dealmaking but not a modest step-up. Three features of the data say more than the count. First, reported deal *value* in these sectors rises, but is concentrated in ownership transfers of existing or distressed assets—the ArcelorMittal Temirtau to Qarmet nationalisation (2023), Eurocopter Kazakhstan and the Lokomotiv rolling-stock plant (re-nationalisations, 2024), and a \$0.8bn 2025 consolidation of rail-freight and forwarding assets—not new productive capacity. Second, greenfield and new-capacity transactions run at one to two per year throughout, with no post-2022 change. Third, across the whole 2015–2025 window the data contain **no** transaction in the specific product lines that dominate the reorientation—a zero that a modest-response CI does not soften.

The result holds across the three databases. Table 3 reports deal counts for the same Kazakhstan universe from each source; coverage differs in level—the transaction-oriented

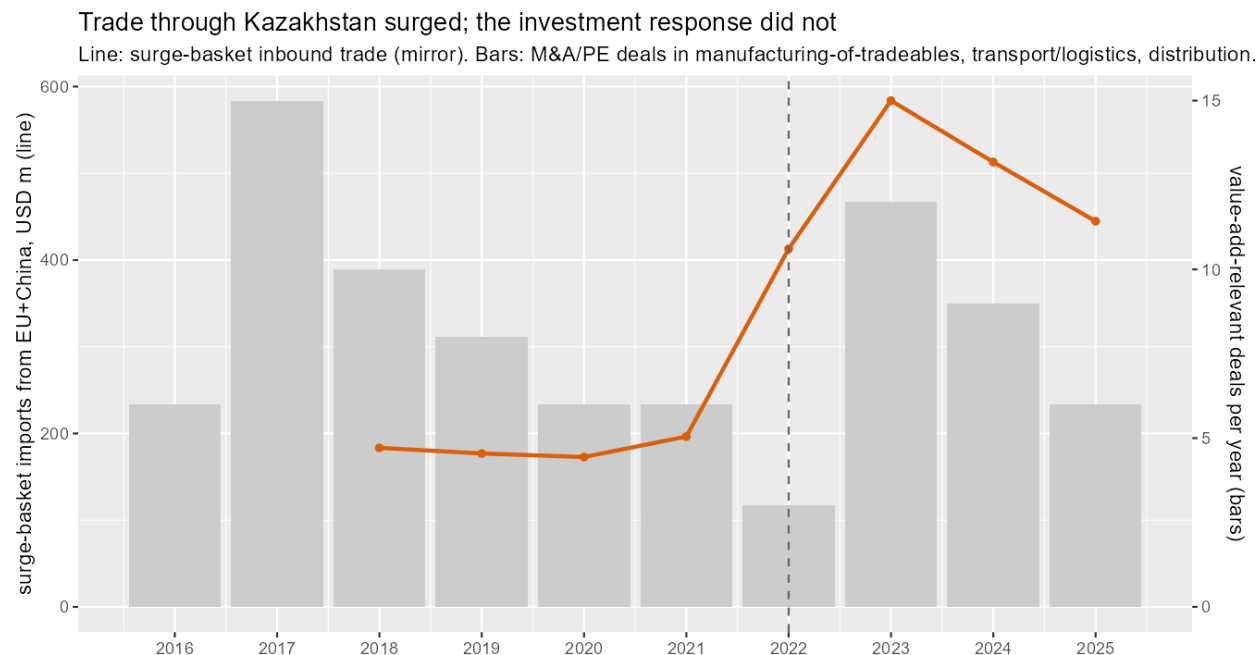


Figure 4: Trade through Kazakhstan surged; the investment response did not. Line: surge-basket inbound trade (mirror), USD million. Bars: annual count of M&A/PE deals in manufacturing of tradeables, transport and logistics, and distribution.

Capital IQ extract records more mid-market M&A, the venture-heavy PitchBook more small early-stage rounds, Preqin fewer of either—but none shows a post-2022 increase in manufacturing or logistics dealmaking. In the Capital IQ extract, which best captures the mid-market industrial deals that a genuine capacity response would generate, the count of manufacturing, transport and distribution deals is essentially unchanged before and after 2022 (5.1 versus 5.2 per year). The FactSet and Dealroom extracts specified in Appendix A are for cross-validation in the final draft.

Kazakhstan is adding vehicle assembly (Chinese marques, KIA, Škoda) and some appliance capacity, but this predates and is largely independent of the reorientation, is kit assembly with low domestic value added, and serves the domestic and Eurasian consumer market rather than the flow of restricted components to Russia. The state investment corporation's post-2022 industrial pipeline is agri-processing and metallurgy, not corridor processing. The one investment signal that is reorientation-linked—the 2025 rail-freight consolidation, with Russia-linked asset names—deepens Kazakhstan's role as a forwarding operation, not a producing one.

Table 3: Deal counts for the Kazakhstan universe, by source and period

Deals per year (unless noted)	Cap. IQ	PitchBk	Preqin	FactSet	Dealroom
All deals, 2015–21	24.0	14.0	3.1	—	—
All deals, 2022–25	23.2	24.0	1.8	—	—
Mfg + transport + distrib., 2015–21	5.1	1.9	0.3	—	—
Mfg + transport + distrib., 2022–25	5.2	1.5	0.5	—	—
Surge-basket lines, 2015–25 (total)	0 ^a	0	0	—	—

^a Two Capital IQ records carry an “electronic” industry tag—a steel-wire-rope manufacturer and a finance holding company—and are misclassifications; there are no transactions in the reorientation’s product lines. FactSet and Dealroom columns pending (Appendix A).

8 Which gate binds?

The framework of Section 3 attributes the null to a closed gate. Which one? The market-access gate is plainly open: under the Eurasian Economic Union the good moves to Russia with no transformation and no border, and the unit-value evidence (Section 6) is consistent with no transformation occurring—so $V_T > 0$ and, with $\mu_P \lesssim \mu_T$, building is dominated in (1) before the option value is added. That alone is close to sufficient for $R \approx 0$. Two within-country comparisons point to the irreversibility gate being independently closed as well. Neither is a clean identification—each holds some things fixed while others move—and we set out their limits.

Captive capital abstained too. Kazakhstan’s state investment corporation (QIC/Baiterek) faces neither an external-finance constraint nor an exit constraint—the institutional gate does not bind for it in the way it binds for an outside equity investor. Yet its post-2022 deployment bypasses the reorientation almost entirely, and so does private capital (Table 4): of the state fund’s 50 industrial and logistics projects financed in 2022–2025, one is in a surge-basket sector and two are in corridor logistics; the rest are domestic-market agri-processing, steel (the Qarmet rescue) and chemicals. This points away from a pure external-finance story: the investor least constrained by capital-market voids also did not build. Two caveats keep it from being decisive. QIC allocates under statutory sectoral programmes rather than scanning the opportunity set, so its silence in component manufacturing is partly a mandate artefact; and a state-owned vehicle is precisely the actor most exposed to secondary-sanctions and reputational risk from financing the rerouting of restricted goods, so its abstention is also consistent with a compliance constraint rather than an irreversibility one. The durable-shock comparison below is what separates these.

A durable shock, the same institutions, a different outcome. Over the same window, Kazakhstan’s *durable* passenger-vehicle demand shock—Western marques withdrawing from Russia, a domestic consumption boom, and Eurasian market access—was met with real

Table 4: Investment 2022–2025 by relation to the reorientation

Relation to the reorientation	State fund (QIC/Baiterek)		Private M&A / PE / VC	
	projects	\$m (fund)	deals	\$m
Surge-basket sector	1	4	1 ^a	10
Corridor logistics / processing	2	117	2	767
Vehicles / machinery (domestic market)	13	1,170	6	138
Unrelated (agri, steel, chemicals, energy)	34	453	192	10,373

Notes. The two dollar columns are on different bases and are not comparable across the divider: “\$m (fund)” is QIC/Baiterek’s own committed amount; “\$m” is total disclosed transaction value, including co-investor and acquirer capital. Deal counts are the comparable quantities. ^a A steel-wire-rope manufacturer carrying an “electronic” industry tag; there is no genuine component-manufacturing transaction. Private-deal values are dominated by oil, mining and banking; the \$767m in corridor logistics is a single 2025 rail-freight consolidation of existing assets.

capacity: a combined Changan/Haval/Chery assembly facility of about 90,000 units, a KIA plant (about \$200m), local assembly of several Škoda models, and a car-multimedia plant in Almaty (2024). The *transitory* component shock was met with no plant at all. We treat this as illustrative rather than as a controlled test, for three reasons. In the deal databases, auto-sector transactions actually *fell* after 2022 (six in 2015–2021, three in 2022–2025); the capacity evidence is from plant announcements and public reporting, not from the deal data, and no symmetric announcement search was run for the component sector. §7 notes that much of the auto capacity predates the reorientation. And the two shocks differ not only in persistence ρ but in sunk-cost intensity, in whether market access requires domestic content (EAEU local-content rules bind for vehicles, not for the re-export), and in the destination market. The comparison is suggestive that a durable, transformation-requiring shock draws capacity where a transitory re-export shock does not; it is not a clean ρ experiment.

The transitory condition is observed, not assumed. The surge-basket flow to Russia peaked in September 2022 and then fell by more than a third over the second half of 2023 (from about \$15m per month in 2023-H1 to about \$9m in 2023-H2) as enforcement tightened (the EU’s eleventh sanctions package and its anti-circumvention instrument, June 2023; sustained bilateral pressure on Kazakhstan through 2023). A firm observing this in real time saw the flow already receding within eighteen months—low ρ , and a σ dominated by policy risk.

The institutional gate explains the residual. It does not explain the shock-specific null—captive capital abstained regardless—but it explains a distinct fact taken up in Section 10: the durable adjacent opportunities the shock reveals, for which the irreversibility gate is open (warehousing and transport support, with a domestic value-added multiplier of 0.82; machinery and fabricated-metal import substitution), are financed only slowly and only by

Table 5: Moderators of the investment response, and Kazakhstan’s values

Moderator	Gate	Kazakhstan, 2022	$R \uparrow$ when
Shock persistence ρ (structural reallocation vs. sanctions-driven)	irreversibility	low (transitory)	high
Uncertainty σ / policy & enforcement risk	irreversibility	high	low
Sunk-cost intensity I of the relevant technology	irreversibility	high (component mfg)	low
Domestic transformation required to reach the market (rules of origin, local content, a border)	market access	none (customs union)	required
Financial depth / exit availability	institutions	shallow, no exit	deep
Private vs. state / directed capital mix	institutions	state-dominated	private-led
Existing productive base / absorptive capacity	institutions	near-zero (electronics output $\approx$ \$230m)	a base exists

directed state capital. The domestic private-equity market has no functioning exit—buy-back is the leading route—capital is dominated by state and development-finance money, and the 2022–25 pipeline is crowded by nationalisations. This is the institutional-voids configuration of Khanna and Palepu [1997] and Rajan and Zingales [1998], and it is the subject of related work.

9 Moderators and external validity

Equation (2) makes the response a function of seven moderators, listed in Table 5 with the value each takes for Kazakhstan. The framework’s testable content is that R is positive only when the irreversibility and market-access gates are both open, and large only when the institutional gate is open as well; a country adverse on the first two should show a null irrespective of its institutions, and a country favourable on all three should show a supply response even with a shallow capital market.

Within the reorientation group. Armenia and the Kyrgyz Republic received the same shock—their surge-basket exports to Russia rose several-fold, breaking in 2022 on the same timing (Section 5)—and share Kazakhstan’s value on every moderator: the shock is sanctions-driven and already receding, enforcement risk is high, and the customs union removes the

need to transform. The framework predicts a null for them too, and the deal data show one: neither records a post-2022 increase in manufacturing or logistics dealmaking, and neither has a domestic component-manufacturing sector to expand.

Contrast cases. Economies that drew capacity investment from a post-2020 tradeable demand shock did so with a favourable configuration on the first two gates, not necessarily the third. Vietnam and Mexico received China-plus-one and nearshoring reallocations widely read as *structural* (high ρ), and in autos the tightening of rules of origin under USMCA made domestic content a condition of market access (a closed market-access gate that forces $V_T \rightarrow 0$). Both invested in capacity despite financial markets that are not uniformly deep. This is the pattern the framework predicts: persistence and a transformation requirement do the work; financial development scales the response but does not gate it.

What we cannot show here. A clean separation of the irreversibility and institutional gates across countries would regress sector-level greenfield investment on shock persistence and financial depth in a panel of intermediaries and contrast cases. In the sample available to us the two are collinear—Kazakhstan, Armenia, the Kyrgyz Republic and the other Caucasus and Central Asian intermediaries are all financially shallow (private credit plus market capitalisation of 24–66% of GDP, all below the comparator median)—and country-level macro aggregates (FDI/GDP, the investment rate, manufacturing value-added growth) are too noisy to isolate a response of this size, and in places point the other way. The within-country comparisons of Section 8 are what the argument leans on; a cross-country panel using greenfield-project data by sector is the natural extension.

10 Where investment would raise value capture

The reorientation is nonetheless informative for investment policy, because the trade data reveal which products have proven demand and proven routes through Kazakhstan. Table 6 sets three quantities side by side for the candidate sectors—the domestic value-added multiplier, the import-substitution gap (imports relative to domestic output), and the post-2022 growth in import demand—after removing the surge-basket lines from the electronics-adjacent chapters; we read them jointly rather than collapsing them into a single index (Figure 5).

Two groups stand out. The **logistics platform**—warehousing and transport support, and wholesale-trade infrastructure—has among the highest domestic value-added multipliers in the economy (0.82) and is the corridor’s binding constraint; investment here creates domestic value and holds it whether or not the Russia flow persists. **Import substitution** in machinery, fabricated metal products and chemicals: Kazakhstan imports about eight

Table 6: Sector characteristics for value-adding investment, Kazakhstan. VA mult. = domestic value-added multiplier $(v'_d L_d)_j$ from the OECD ICIO block; Output = gross output, \$m; Imp./output = 2023 imports over domestic output; Imp. growth = ratio of 2023 to 2019 import value. Dashes: not applicable (platform sectors are not import-competing) or not computed. Surge-basket HS6 are netted out of the electronics and machinery chapters.

	Sector	VA mult.	Output \$m	Imp./ output	Imp. growth
Platform	Warehousing & transport support	0.82	6,800	—	—
	Wholesale & retail trade infrastr.	0.82	81,000	—	—
Substitution	Machinery & equipment n.e.c.	0.80	1,160	7.9	1.27
	Fabricated metal products	0.81	1,850	1.8	1.22
	Chemicals	0.79	3,480	1.7	1.52
Avoid	Computer, electronic & optical (surge basket)	0.69	230	—	—
	Motor vehicles & parts (kit assembly)	0.69	1,530	6.1	2.55

times what it produces in machinery and roughly twice its output in the other two, the domestic value-added multiplier is around 0.80, and a nascent deal base exists. By contrast, localising the surge-basket products is the worst option—computer, electronic and optical manufacturing has the *lowest* value-added multiplier in the economy (0.69) and a near-zero domestic base—and the same is true of motor-vehicle kit assembly.

These opportunities are investable at scale only if the state investment corporation shifts from lead direct investor to co-investor behind private leads, with larger syndicated tickets and a credible exit—trade sale or listing—rather than the buy-back that currently dominates. That is a question about capital-market institutions, taken up in related work.

11 Discussion

Kazakhstan’s trade with Russia and with the West in these products moved by an order of magnitude after 2022. The domestic economic content of that movement—value added about a tenth of the manufacturing equivalent, customs duty under one percent of revenue, a fraction of one percent of GDP, and no detectable investment response—is small. The country is functioning as a corridor, not a site of production.

The result speaks to a general question: when does a trade shock industrialize a country, and when does it merely pass through? The framework of Section 3 gives a three-part answer. Production is a plausible response only when the demand cannot be met by trade



Figure 5: Where investment buys the most domestic value added: domestic value-added multiplier against import-substitution headroom (imports 2023 / domestic output, log scale), bubble size = domestic output. The surge-basket products (computer/electronic/optical) sit at the bottom.

alone (a closed market-access gate) and only when the shock is durable and certain enough, and the technology cheap enough to sink, that building beats waiting (an open irreversibility gate); and it is actually realised only where the capital market can fund the projects that pass (an open institutional gate). Kazakhstan is adverse on the first two: the customs union makes trade a close substitute for production, so the shock is largely absorbed into the trade channel, and the flow is transitory and policy-contingent, so even the un-absorbed demand does not clear the investment threshold. That the state investment corporation, which does not face the capital-market constraint, also did not build points to the block being upstream of finance, though its mandate and its own sanctions exposure cloud the comparison. The institutional gate then explains the missed alternative: the durable opportunities the shock reveals, in logistics and machinery, are left to slow, directed state capital. The same configuration holds for the other reorientation intermediaries—Armenia and the Kyrgyz Republic—and the framework’s contrast cases (structural reallocations, or rules-of-origin regimes that force domestic content) are where a supply response does occur, even without a deep capital market. For the design of the restrictions the reorientation responds

to, the implication is that the intermediary retains little, so that the economic pressure of enforcement falls on the final buyer and the original seller rather than on the transit country. For Kazakhstan itself, the static value that is missed is the smaller loss: an open market-access gate also forgoes the technology transfer and learning that a rules-of-origin regime would have forced from foreign suppliers [de Souza et al., 2026], which is where the long-run cost of being a corridor sits.

Limitations. The argument for which gate binds rests on two within-country comparisons (captive versus constrained capital, and a durable versus a transitory shock) rather than on a cross-country panel, and neither is a clean experiment; the panel that would separate the irreversibility and institutional gates directly requires greenfield-project data by sector and country and is left for future work. The investment null is *over*-determined—the 2022 unrest, secondary-sanctions exposure and the nationalisation wave would each depress entry—so we can bound its magnitude (Section 7) but not attribute it to the trade shock alone. On measurement: the inbound side is partner-reported (mirror) data, which overstates Kazakh absorption for onward-moving goods; the inbound flow is dominated by a large pre-existing China component, so we measure value capture on the incremental Western flow and read the West-plus-China inbound only at the product level (where line and year fixed effects absorb the China trend), not in the aggregate; the surge basket is selected on the post-2022 outcome, and although a pre-specified priority-list version and a randomisation-inference benchmark give the same result, the cross-sectional difference-in-differences should be read alongside the structural breaks and the neighbour comparison rather than on its own; a single input–output vintage (2019) and the OECD ICIO’s relatively high domestic-content shares for Kazakhstan; the retained trade margin bounded between a national-accounts convention (6–14%) and a freight-netted matched-cell estimate ($\approx 12\%$), not estimated line by line; 2024 partial in the monthly data; and the HS-to-industry crosswalk in Section 10, which is approximate at the chapter level. The FactSet and Dealroom columns of Table 3 are pending.

A Deal-database query specification

The same Kazakhstan deal universe is extracted from up to five commercial databases; the analysis in this draft uses the first three (Capital IQ, PitchBook, Preqin), with the FactSet and Dealroom extracts specified here and run for the final draft. Common parameters: *target/issuer geography* = Kazakhstan (country of incorporation or primary operations); *announced date* between 1 January 2015 and 31 December 2025; *deal types* =

merger/acquisition, majority/minority stake, private-equity buyout and growth, venture (all stages), joint venture, and PIPE/private placement; *status* = completed or announced (pending and withdrawn recorded separately). Financials in USD at announcement.

Capital IQ (S&P): Screening > Transactions; Target Company Location = Kazakhstan; Transaction Types as above; Announced Date range as above. Export all columns; extraction date recorded in the replication file.

PitchBook: Advanced Search > Deals; Company HQ Location = Kazakhstan; Deal Types = M&A, Buyout/LBO, PE Growth/Expansion, all VC stages, Angel, Accelerator/Incubator; Deal Date range as above.

Preqin: Private Equity > Deals & Exits; Portfolio Company Country = Kazakhstan; Deal Date range as above.

FactSet: Mergers & Acquisitions / Private Markets; Target Nation = Kazakhstan; Transaction Type = as above; Announce Date range as above. *[To run.]*

Dealroom: Companies > HQ country = Kazakhstan; Funding rounds and M&A; Round date range as above; include grants and ICO/token as separate flags. *[To run.]*

De-duplication is on target name, announced date (± 30 days) and deal type. Sector classification is harmonised to ISIC rev. 4 divisions from each database’s native taxonomy via a documented crosswalk. All five extracts, the crosswalk, and the reconciliation script are in the replication package.

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